Math 133 **Name (Print):** _____

Fall 2026

Midterm 1

October 2, 2026

Time Limit: 50 minutes

This exam contains 6 pages (including this cover page), 5 problems, and 50 points. Check that you have all pages. Write your name above and your initials on each remaining page.

This exam is closed book and closed notes. You may use a scientific calculator if you wish. Graphing calculators, phones, computers, and other electronic devices are not allowed. No reference sheets or outside resources are permitted.

- Show your work. Partial credit may be given for correct reasoning; unsupported answers may not receive full credit.
- Simplify your answers. Give exact values unless a decimal approximation is requested.
- Include units in business applications where appropriate.
- Write your solutions in the spaces provided. If you need scratch paper, ask for it and transfer your final work to the appropriate exam page.

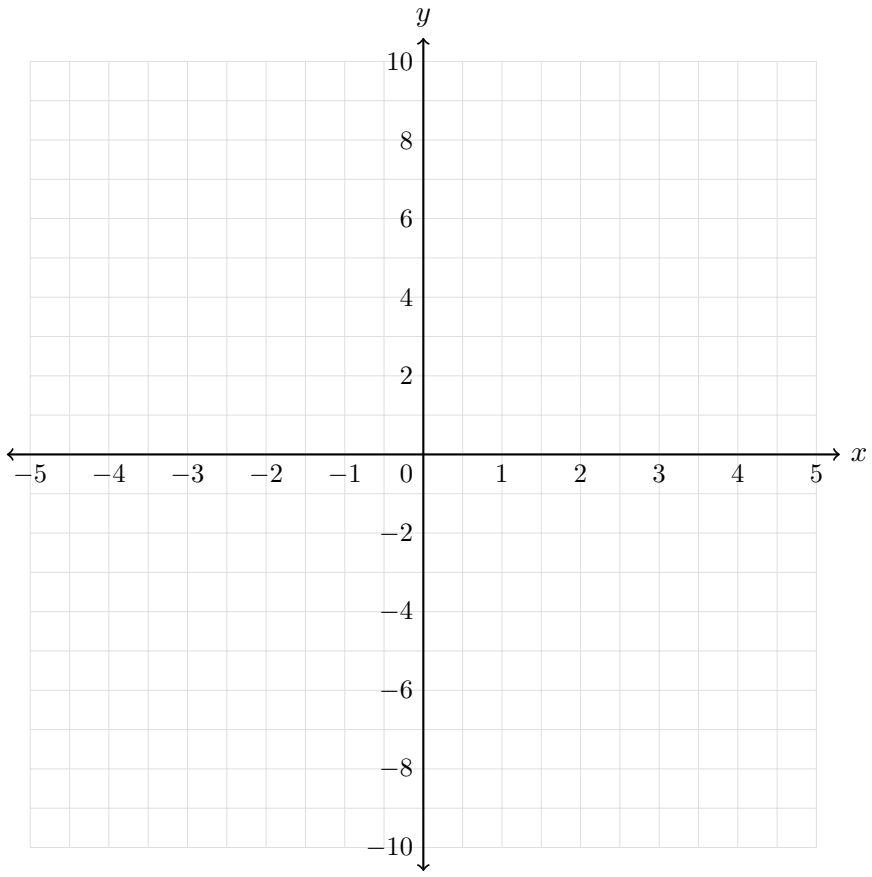
Question	Points	Score
1	12	
2	9	
3	12	
4	5	
5	12	
Total:	50	

1. (12 points) Let $f(x) = x^2 - 2x - 8$.

   (a) Solve $f(x) = 0$.

   $x =$ _____

   (b) Sketch the graph of f on the grid. Label the vertex with its coordinates.

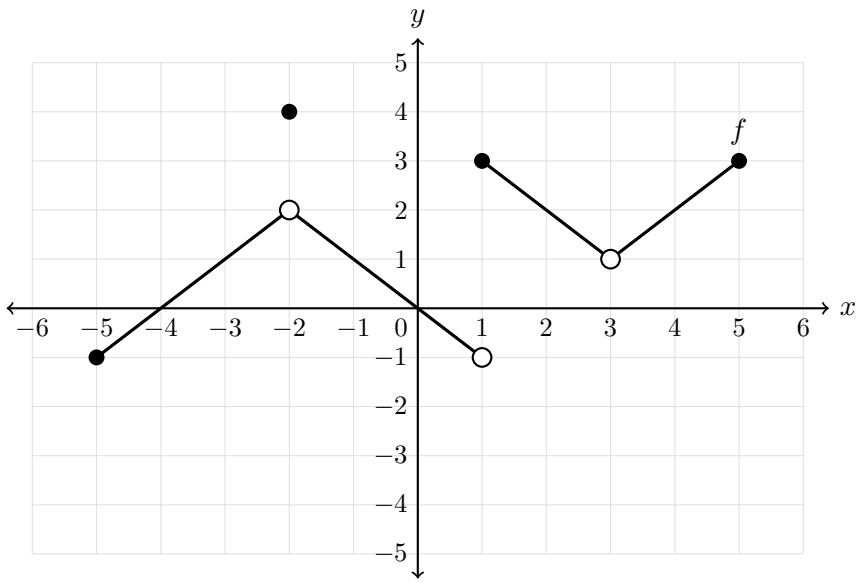


   (c) Find the maximum and minimum values of f on $[0, 5]$, and give all x-values where each occurs.

   Maximum: _____ at $x =$ _____

   Minimum: _____ at $x =$ _____

2. (9 points) The entire graph of f is shown below. The pieces are line segments. Open circles are excluded; filled circles are included.



(a) Evaluate the function at the following values, or write UND if the function is undefined at the value.

$f(-3) =$ _____ $f(-2) =$ _____ $f(0) =$ _____ $f(1) =$ _____ $f(3) =$ _____

(b) Evaluate each limit. Write DNE if a limit does not exist.

$\lim_{x\to -2^-} f(x) =$ _____ $\lim_{x\to -2^+} f(x) =$ _____ $\lim_{x\to -2} f(x) =$ _____

$\lim_{x\to 1^-} f(x) =$ _____ $\lim_{x\to 1^+} f(x) =$ _____ $\lim_{x\to 1} f(x) =$ _____

$\lim_{x\to 3^-} f(x) =$ _____ $\lim_{x\to 3^+} f(x) =$ _____ $\lim_{x\to 3} f(x) =$ _____

(c) Give the open intervals on which f is increasing and those on which it is decreasing.

Increasing: _____

Decreasing: _____

3. (12 points) A moving company handles jobs of 1 to 20 boxes. For a job of that moves fewer than 11 boxes, it hires one mover; for a job of 11 to 20 boxes, it hires two movers. Each mover is paid \$40 per job. Packing materials cost an additional \$2 per box. These are the company's only costs. Let q be the number of boxes in a job.

   (a) Write a piecewise formula for the total cost $C(q)$, in dollars, including the allowed values of q for each piece. Make sure you specify the exact domain on which your function is defined.

$$C(q) = \begin{cases} _____ & \text{if } _____, \\ \\ _____ & \text{if } _____. \end{cases}$$

   (b) Find the marginal cost of the 10th box and of the 11th box, meaning the exact additional cost of the 10th and of the 11th box respectively. Include units.

Marginal cost of 10th box: _____ Marginal cost of 11th box: _____

(c) The company's revenue for a job of q boxes is

$$R(q) = 20q - \tfrac{1}{2}q^2 \quad \text{dollars.}$$

Find the average rate of change of revenue from 10 to 14 boxes. Include units.

Average rate of change of revenue from 10 to 14 boxes: _____

4. (5 points) Evaluate each limit.

(a) $\lim_{x\to 1} \dfrac{x^2-9}{x-3}$

$$\lim_{x\to 1} \frac{x^2-9}{x-3} = _____$$

(b) $\lim_{x\to 3} \dfrac{x^2-9}{x-3}$

$$\lim_{x\to 3} \frac{x^2-9}{x-3} = _____$$

5. (12 points) A machine's resale value is modeled by

$$V(t) = 1000(0.90)^t, \qquad t \geq 0,$$

where t is years after purchase and $V(t)$ is in dollars.

(a) State the machine's initial value and the percentage of its remaining value that it loses each year.

Initial value: _____ Annual percentage loss: _____

(b) Find the dollar loss during the first year and the dollar loss during the second year. Does the model predict the same dollar loss each year?

First-year dollar loss: _____ Second-year dollar loss: _____

Same dollar loss each year? **Yes** **No** (Circle one)

(c) For which times $t \geq 0$ is the machine worth less than \$500? Give an exact answer.

Times: _____